

ETHIOPIAN HEALTH PROFESSIONALS LICENSING EXAMINATION(EHPLE) PRACTICE TEST BOOKLET 2025

MEDICAL RADIOLOGY TECHNOLOGY



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MINISTRY OF HEALTH-ETHIOPIA

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HEALTHIER CITIZENS FOR PROSPEROUS NATION!



Message from the State Minister, Ministry of Health – Ethiopia



Mrs. Frehiwot Abebe

Improving the quality of healthcare remains a critical global priority and is essential for achieving sustainable development and universal health coverage. One of the most effective ways to ensure high standards in healthcare delivery is through the consistent assessment of professional competencies. In line with this commitment, the Ministry of Health of Ethiopia launched the Ethiopian Health Professionals Licensing Examination (EHPLE) in 2019 for undergraduates in seven core health disciplines. Since its inception, the program has expanded to encompass 17 health professions, reflecting our continued effort to build a competent and responsive health workforce.

The EHPLE was established with the primary aim of identifying and certifying health professionals who possess the minimum competencies required to provide safe and effective care. This ensures that all practicing professionals are equipped to meet the expectations of the communities they serve. The exam is coordinated by the Health and Health-Related Institutions and Professionals' Regulatory Lead Executive Office (LEO), which consists of four specialized desks. The LEO plays a vital role in system strengthening, enabling the exam to be conducted with increased efficiency, organization, and quality assurance.

It is important to recognize that the EHPLE is not a conventional academic or recruitment examination. Rather, it is a rigorous, competency-based assessment designed to evaluate the practical skills and critical thinking abilities that are essential for professional practice. This *Practice Items Booklet* is part of our broader strategy to support examinees and educators in preparing for the licensing process. It offers a realistic overview of the exam structure, promotes a deeper understanding of its objectives, and fosters transparency and sustainability in implementation.

The development of this booklet has been made possible through the collaborative efforts of experts from higher education institutions, the Addis Ababa University Testing Center, and professional associations. Their insights and dedication have been instrumental in shaping this resource. I extend my heartfelt thanks to all contributors and to the Ministry of Health team for their tireless work and steadfast commitment to excellence in healthcare regulation.

We believe this booklet will serve as a valuable tool for candidates preparing for the examination and for educators guiding them toward success. Together, we are building a stronger, more competent health workforce for Ethiopia.

Introduction EHPLE

Welcome to the 2025 Edition of the Ethiopian Health Professionals Licensing Examination Practice Test Booklet.

This resource is designed to support candidates in their preparation for the national licensing examination, ensuring they are well-equipped with the knowledge and clinical reasoning skills required to deliver competent and ethical healthcare across Ethiopia.

In Ethiopia, the licensing examination serves as a pivotal tool in evaluating the competency of health professionals who have completed their academic training and are ready to transition into independent practice. The exam not only ensures public safety by verifying professional readiness but also upholds national and international standards in healthcare service delivery. As the healthcare system continues to expand and evolve, a strong, well-regulated professional workforce becomes even more essential.

This practice booklet contains 200 multiple-choice questions modelled after the structure, content, and format of the actual licensing examination. The questions span across core clinical domains essential for entry-level practice. This includes areas such as Patient care, leadership and management, ethics, and Health Promotion as outlined in the Information Booklet 2024.

The booklet is intended to:

- Familiarize candidates with the structure and cognitive level of the actual licensing examination.
- Offer opportunities for self-assessment and identification of strengths and gaps in knowledge.
- Encourage critical thinking and clinical decision-making in scenarios reflective of real-world healthcare practice in Ethiopia.
- Reduce exam-related anxiety by providing realistic practice experiences.

Administered under the auspices of the Ministry of Health and relevant regulatory bodies, the Ethiopian Health Professionals Licensing Examination is grounded in evidence-based frameworks and aligned with both national needs and global best practices. It has been structured to assess a range of competencies, including medical knowledge, communication, professionalism, and safe clinical practice.

This practice test is not a predictive tool, and performance on it does not guarantee similar outcomes on the actual exam. However, it provides an invaluable opportunity for familiarization with question styles, pacing strategies, and content review. We strongly recommend that candidates simulate real testing conditions while using this resource, including timing their responses and reviewing rationale for both correct and incorrect answers.

The goal of licensing examinations is not only to regulate entry into the profession but also to inspire excellence and lifelong learning. By engaging with this practice test, you are taking a proactive step toward professional development, accountability, and a commitment to serving your patients with competence and compassion.

We wish you success on your licensing journey and beyond.

Acknowledgements

This *Practice Test Items Booklet for the Ethiopian Health Professionals Licensing Examination* is the result of a collaborative effort among dedicated professionals committed to advancing the quality of health sciences education in Ethiopia. The Ministry of Health extends its sincere appreciation to all those who contributed their time, expertise, and insights to the development of this essential resource.

Special recognition is given to the item developers and reviewers from various higher education institutions, both public and private, whose academic rigor and practical experience were instrumental in shaping the content and ensuring its relevance to real-world healthcare practice. Their contributions have helped create a high-quality, standardized practice tool that reflects national competencies and global benchmarks.

We are especially grateful to the team at the Addis Ababa University Testing Center, whose technical support and academic guidance were critical throughout the process. Their leadership in assessment design and educational evaluation has significantly strengthened the quality of this booklet.

We also thank the committed staff of the Ministry of Health, particularly those within the Health and Health-Related Institutions and Professionals' Regulatory Lead Executive Office (HHRIPR LEOs), whose coordination, oversight, and unwavering dedication ensured the successful completion of this project.

Together, these collective efforts reflect our shared vision of building a competent, ethical, and resilient health workforce for Ethiopia.

Medical Radiology Technology

1. A radiology technologist performed a PA abdominal x-ray on a 12-year-old patient and wanted to use a device to indicate the location of the radiation field and control its size. What is the most appropriate equipment to be applied in this case?
A. Filter
B. Gird
C. Collimator
D. Control console
ANSWER: C
2. A radiology technologist wanted to perform a chest x-ray using a mobile x-ray unit on a young patient in the intensive care unit with the lowest possible occupational radiation exposure. What is the most appropriate radiation protection technique to be applied in this case?
A. Increasing tube current
B. Shielding x-ray equipment
C. Decreasing the distance from the source
D. Increasing the distance from the source
ANSWER: D
3. A medical radiology technologist processed a film in a dark room after performing a lumbar x-ray. Upon viewing the film, he sensed that the energy of the X-ray beam had been reduced, as the image appeared to be foggy. What is the most likely exposure factor causing the problem?
A. Time
B. Distance
C. Milliamperage
D. Peak kilovolts
ANSWER: D
4. An adult patient with a history of a motorcycle accident visited a radiology department for a shoulder X-ray. Antero-posterior projection was performed, and the physician noticed a posterior dislocation and recommended additional projection to be taken to rule out posterior dislocation of the humerus from the gleno-humeral joint. What is the most appropriate radiographic projection to be applied in this case?
A. Oblique
B. Lateral
C. Supero-inferior
D. Postero-anterior
ANSWER: C
5. A 25-year-old female patient sustained a car accident while she was driving without her seat belt on. After the collision, she started to feel severe pain in her mid-right clavicle area, for which X-ray was ordered. The anteroposterior projection was able to demonstrate a possible fracture. However, the technologist decided to add an alternative projection to confirm the fracture. What is the most appropriate projection that should be added in this case?
A. Infero-suprior
B. Lateral deubitus
C. Posteio-anterior oblique
D. Anteio-posterior oblique
ANSWER: A

6. A radiology technologist wants to perform a true anteroposterior shoulder joint x-ray for a patient with suspected shoulder joint dislocation and proximal humeral fracture. The patient feels severe pain and is unable to maintain the desired position. What is the most appropriate measure the technologist should take?
- A. Not to rotate the arm
 - B. To use restraints to maintain position
 - C. To use a longer exposure time and high KV
 - D. To instruct the patient to overcome the pain
- ANSWER: A
7. A 58-year-old female patient came to the department of radiology from the emergency unit after a falling down accident onto the outstretched hands. This trauma resulted in a fracture of the shaft of the humerus, which was confirmed on her shoulder x-ray radiograph. She has pain, bruising, and shows signs of wrist drop. What is the most likely explanation of the patient's condition in this case?
- A. Dislocation
 - B. Impingement
 - C. Vascular injury
 - D. Radial nerve damage
- ANSWER: D
8. A patient with a history of a car accident came to the radiology department for upper arm x-ray to rule out humeral neck fracture. Clinical history and physical examination indicate no open fracture except pain around the shoulder. The technologist positioned the patient by applying proper collimation during the projection of the anteroposterior. What is the most likely measure that the technologist applied in this case?
- A. Included the coracoid process and distal end of the humerus
 - B. Included both articular ends of the humerus and the clavicle
 - C. Included the acromion process and proximal half of the humerus
 - D. Included the medial border of the scapula and both articular ends of the humerus
- ANSWER: C
9. A male patient with a history of trauma to his elbow was sent to an X-ray unit. On his AP and lateral views, the technologist noticed a radial head and neck fracture but was not able to conclude its extent and the presence of any further dislocation. Thus, the technologist decided to add an alternative projection. What is the most appropriate additional view needed in this case?
- A. AP oblique with internal rotation
 - B. PA with caudal tube angulation
 - C. PA with cephalic tube angulation
 - D. AP oblique with external rotation
- ANSWER: D
10. A patient from an outpatient department was sent to a radiology department for a wrist joint x-ray to rule out carpal tunnel syndrome. The technologist performed the routine projections and noticed abnormal calcification, fracture, and bony changes in the carpal sulcus. Hence, decided to take an additional view of the wrist. What is the most appropriate projection to be added in this case?
- A. Tangential-inferio-supeiror
 - B. Lateral oblique
 - C. Medial oblique
 - D. Antero posterior
- ANSWER: A

11. A patient with complaints of thumb pain and swelling at the first metacarpal after a falling down accident came to the x-ray unit. The physician has suspected injury to the ulnar collateral ligament of the thumb as a result of hyperextension of the thumb.
What is the most appropriate projection to evaluate this case?
- A. Lateral
 - B. Oblique
 - C. AP axial
 - D. PA stress
- ANSWER: D
12. A 67-year-old male patient came to a radiology unit with chief complaints of fever, general malaise, weight loss, and weakness of the knee joints. The bilateral AP and lateral knee joint radiograph shows local osteopenia, fusiform soft tissue swelling, marginal and central bone erosions with joint-space widening of the bilateral knee joints.
What is the most likely diagnosis in this case?
- A. Osteoarthritis
 - B. Osteomyelitis
 - C. Paget's disease
 - D. Rheumatoid arthritis
- ANSWER: D
13. A patient with a history of trauma to the knee joint came to an emergency OPD. Then the physician requested a knee joint x-ray to rule out a patellar fracture. After the radiology technologist performed the basic AP and lateral projections, the patellar fracture was not diagnosed. An additional image with alternative projection was required to assess the retro-patellar joint space and confirm if there is any vertical patellar fracture.
What is the most appropriate additional projection required in this case?
- A. PA-oblique projection
 - B. Standing projection
 - C. Skyline projection
 - D. Stress projection
- ANSWER: C
14. A 28-year-old male patient with a history of car accident came to the radiology unit for x-ray examination of the lower leg. The physician suspected fracture of the distal fibula. The radiology technologist prepared the X-ray machine to perform the requested examination.
What is the most important technique to be used in this case?
- A. Include the ankle joint
 - B. Focus on the area of fracture
 - C. Knee and ankle joints must be included
 - D. Include the lower third of the lower leg
- ANSWER: C
15. A 75-year-old male patient with a complaint of pain while walking had a foot x-ray done. On the radiograph, the radiology technologist noticed that joint spaces are narrowed and are irregular with sclerotic articular surfaces and spurs.
What is the most likely diagnosis?
- A. Osteoarthritis
 - B. Osteomalacia
 - C. Osteoclastoma
 - D. Osteoid osteoma
- ANSWER: A

16. A 35-year-old male patient with a history of foot pain came to a radiology unit for x-ray examination. A radiology technologist positioned the patient for a foot x-ray with basic projection.

What is the most appropriate patient position in this case?

- A. Dorsi-plantar and Lateral
- B. Dorsi-plantar oblique and Lateral
- C. Dorsi-plantar -erect and Lateral- erect
- D. Dorsi-plantar and Dorsi-plantar oblique

ANSWER: D

17. A 33-year-old female patient with a complaint of severe pain on her calcaneus came to the radiology unit for x-ray examination. The physician wanted to view the talocalcaneal joint. The radiology technologist performed a lateral-mediolateral projection to demonstrate the requested examination and wanted to make sure that proper positioning was applied upon viewing the radiograph.

What is the most appropriate explanation for this case?

- A. Proximal one-third of the lower leg is being viewed
- B. Tibiotalar joint being closed with uniform joint space
- C. Lateral malleolus being superimposed over distal tibia and talus
- D. Distal fibula superimposed by distal tibia, talus, and calcaneous

ANSWER: D

18. A 28-year-old male patient diagnosed with diabetes presented to an X-ray room for lower leg x-ray examination with chief complaints of fever, severe pain associated with malaise, local erythema, and oedema on the right leg. An AP and Lateral right leg radiograph after six weeks shows abnormal trabeculation and lysis within the distal tibial metaphysis with associated periosteal reaction.

What is the most likely diagnosis for this patient?

- A. Osteoarthritis
- B. Osteomyelitis
- C. Perthes disease
- D. Psoriatic arthropathy

ANSWER: B

19. A 40-year-old male patient with a complaint of foot pain and difficulty standing straight came to the radiology unit for x-ray examination of the foot. The physician requested to demonstrate the bones of the feet to show the condition of the longitudinal arches while standing upright. The technologist positioned the patient erect on a digital IR, with full body evenly distributed on both feet and directed straight ahead, parallel to each other. Then he directed CR 15 degree posteriorly to the midpoint between the feet at the level of the base of the metatarsals.

What is the most likely projection performed in this case?

- A. Lateral- mediolateral
- B. Antero-Posterior mortise
- C. Postero-Anterior tangential
- D. Antero-Posterior weight bearing

ANSWER: D

20. A 67-year-old male patient came to x-ray room with complaints of severe pain, ankle joint swelling, and high sensitivity while being contacted. His laboratory result showed increased level of uric acid. Upon interpreting the radiograph, the radiology technologist demonstrated eccentric peri-articular 'punched out' erosions with uniform joint-space narrowing.
What is the most likely diagnosis for this case?
A. Osteomyelitis
B. Gout arthritis
C. Osteonecrosis
D. Ankylosing spondylitis
ANSWER: B
21. A 56-year-old female patient came to a radiology department for a cervical X-ray with a known history of rheumatoid arthritis.
What is the most appropriate projection in this case?
A. Antero-Posterior Open mouth
B. Postero-Anterior oblique
C. Lateral flexion-extension
D. Wagging jaw
ANSWER: C
22. An elderly male patient, with complaints of back pain and difficulty bending down, was referred to an X-ray department for Lumbosacral and sacroiliac X-ray. The radiograph showed asymmetrical spurs on many vertebrae, making it look like a bamboo with a hazy and fused sacroiliac joint.
What is the most likely diagnosis for this patient?
A. Spodylosis
B. Spondylisis
C. Spondylolisthesis
D. Ankylosing spondylitis
ANSWER: D
23. A 10-year-old female patient who has scoliosis came to a radiology department for a lumbar x-ray. The radiology technologist performed the examination using routine projection. However, the technologist planned to add additional projection to differentiate the deforming curve from the compensatory curve.
What is the most appropriate projection required in this case?
A. Anterior posterior axial
B. Lateral-hyperextension
C. Antero-Posterior ferguson method
D. Posterior -anterior right and left bending
ANSWER: C
24. A 37-year-old patient with a history of back pain came to a radiology department for lumbar x-ray to assess the defects of the pars interarticularis. The radiology technologist received the request and performed the examination using appropriate positioning.
What is the most likely positioning applied in this case?
A. Lateral-hyperextension
B. Left and right posterior oblique
C. Antero-Posterior Ferguson method
D. Lateral-hyperextension and hyperflexion
ANSWER: B

25. A 37-year-old patient came to a radiology department on a trolley and cervical collar for cervical x-ray with a history of falling down accident from a building. A radiology technologist received this patient and positioned the patient. What is the most appropriate projection for this patient?
- A. Lateral erect
 - B. Swimmer's lateral
 - C. Lateral decubitus
 - D. Flexion and extension lateral
- ANSWER: C
26. A radiology technologist performed a lateral lumbar x-ray. However, the technologist could not demonstrate a clear intervertebral disc space on the radiograph. What is the most likely positioning fault in this case?
- A. An exaggerated collimation.
 - B. Lack of suspended expiration
 - C. Misuse of Left or right marker
 - D. The spine is not parallel with the cassette
- ANSWER: D
27. A 57-year-old male came to a radiology department for sacroiliac joint x-ray examination with complaints of the right side of his hip is lower than his left one, minor pain, and discomfort while walking. The radiology technologist assisted the patient to lie in supine with arms at side and head on a pillow and rotated the patient's body to perform posterior oblique projections. What is the most appropriate degree of rotation required in this position?
- A. 25 to 30
 - B. 20 to 25
 - C. 30 to 35
 - D. 15 to 20
- ANSWER: A
28. A 75-year-old patient with a complaint of severe abdominal pain came to an x-ray room for abdominal x-ray. The radiology technologist performed the examination and demonstrated medium opacity band in the paracolic gutters, loss of clarity of the liver edge, and medial displacement of the ascending colon on the radiograph. What is the most likely diagnosis for this case?
- A. Ascites
 - B. Perforation
 - C. Appendicitis
 - D. Obstruction
- ANSWER: A
29. A 28-year-old patient came to a radiology department for abdominal x-ray exam to assess the air-fluid level. The radiology technologist positioned the patient for an erect anterior-posterior projection. The technologist instructs the patient about breathing during exposure. What is the most appropriate breathing instruction required in this case?
- A. Expose at the end of inspiration
 - B. Expose at the end of expiration
 - C. Expose at the beginning of expiration
 - D. Expose at the beginning of inspiration
- ANSWER: B

30. A 24-year-old female patient with a history of stab injury to the abdomen came to an x-ray room on a stretcher for abdominal x-ray examination to rule out abdominal free air collection. What is the most appropriate patient position to be applied in this case?
- A. Anterio posterior
 - B. Posterio anterior
 - C. Anterior oblique
 - D. lateral decubitus
- ANSWER: D
31. A 26-year-old male patient came to an x-ray room with a history of colicky abdominal pain and vomiting. The plain abdominal x-ray image of this patient showed multiple dilated loops of bowel within the central abdomen. What is the most likely diagnosis for this patient?
- A. Cecal volvulus
 - B. Chilaiditi's sign
 - C. Large bowel obstruction
 - D. Small bowel obstruction
- ANSWER: D
32. A male patient with abdominal x-ray request paper came to a radiology department. The radiology technologist positioned the patient stand against the upright bucky and arms at the side, away from the body. Then, the technologist adjusted the height of the image receptor so that the center is approximately five centimeters above the iliac crest, and then collimated the entire abdomen properly. What is the most likely positioning applied in this case?
- A. Lateral erect
 - B. Oblique erect
 - C. Postero-Anterior erect
 - D. Antero-Posterior erect
- ANSWER: D
33. A radiology technologist performed an AP axial skull x-ray. On the image, the technologist observed two anatomical structures within the shadow of the foramen magnum. What is the most likely anatomy demonstrated in this case?
- A. Mandibular condyle & Clivus
 - B. Occipital bone and Petrous ridge
 - C. Coronal suture & Mastoid air cell
 - D. Dorsum selle & Posterior clinoid process
- ANSWER: D
34. A male patient came to a radiology department for a skull x-ray. A radiology technologist positioned the patient. The patient was placed in a semi-prone position on the table, the mid-sagittal plane was made parallel with the detector plane, and the interpupillary line was adjusted perpendicular to the detector plane. What is the most likely patient positioning applied in this case?
- A. AP
 - B. Lateral
 - C. PA (Caldwell)
 - D. Submentovertical
- ANSWER: B

35. A well-defined, radio-dense structure arising from the outer table of the skull was identified in the frontal sinus when the technologist interpreted a skull radiograph.
What is the most likely diagnosis for this case?
- A. Osteomas
 - B. Meningioma
 - C. Paget's disease
 - D. Fibrous dysplasia
- ANSWER: A
36. A radiology technologist identified a homogeneous and radio-dense structure in the posterior part of the submentovertical skull radiograph in the lower area of the film.
What is the most likely anatomy demonstrated in this case?
- A. Vomer
 - B. Occipital bone
 - C. Petrous pyramid
 - D. Mandibular ramus
- ANSWER: B
37. A patient with complaints of nasal stiffness, headache during cold weather, and bad smell came to a radiology department for facial x-ray. A technologist made the patient sit facing the upright grid device, the midsagittal plane parallel with the long axis of the upright grid and perpendicular to the film plane. The film plane formed 37 degree to the orbitomeatal line and was perpendicular to the mentomeatal line during head positioning. The central ray was then directed perpendicular to the film, exiting at the acanthion.
What is the most likely projection applied in this case?
- A. PA paranasal sinus
 - B. Lateral paranasal sinus
 - C. Parietoacanthial paranasal sinus
 - D. Submentovertical paranasal sinus
- ANSWER: C
38. A male patient came to the department of radiology for a facial X-ray to check a mandibular ramus fracture. The radiology technologist on duty positioned the patient and performed an examination in the PA position and directed the central ray perpendicular to the acanthion. After the examination, the technologist assessed the image and observed that both orbits were filled with radio-dense structures.
What is the most likely anatomy identified in this case?
- A. Maxilla
 - B. Petrous ridge
 - C. Sphenoid bone
 - D. Zygomatic bone
- ANSWER: B
39. A chest x-ray film interpretation report of a young male patient, made by a radiology technologist working in a primary hospital, shows that the apex of the heart is toward the right side and the fundal gas is viewed under the right hemidiaphragm.
What is the most likely diagnosis for this patient?
- A. Hiatus hernia
 - B. Situs inversus
 - C. Cardiac transplant
 - D. Ventricular aneurysm
- ANSWER: B

40. A 25-year-old male patient came to a radiology unit for a chest x-ray as requested by the immigration center for his visa approval. His radiograph shows a specific ridge in the inside portion of the trachea, where it divides into right and left bronchi. What is the most appropriate name of this anatomical structure?
- A. Hilum
 - B. Pleura
 - C. Carina
 - D. Alveoli
- ANSWER: C
41. A 14-year-old male patient came to a radiology department with a chest x-ray request paper. The radiographer performed a PA chest x-ray. Upon viewing the radiograph, the distance between the left sternal ends of the clavicles to the central line of the spine is wider than the right side. What is the most likely positioning error made in this case?
- A. Patient rotation
 - B. Improper central point
 - C. Depressed patient's chin
 - D. Depressed patient's shoulder
- ANSWER: A
42. A radiology technologist noticed only four complete anterior ribs and seven posterior ribs upon assessing a chest radiograph of a young male patient. What is the most likely explanation for this finding?
- A. Hyper-expanded lung
 - B. Rotation of the patient
 - C. Poor inspiratory effect
 - D. Under-penetrated film
- ANSWER: C
43. A 28-year-old patient came to X-ray department for chest x-ray. On his prescription paper, the physician specifically requested a chest x-ray to better visualize the size and contours of the heart and great vessels. What is the most appropriate radiographic projection in this case?
- A. PA
 - B. AP
 - C. Lateral
 - D. Oblique
- ANSWER: D
44. A 32-year-old female patient, with a history of a motor bicycle accident, was brought to a radiology unit by a wheelchair for chest x-ray due to her inability to stand or walk. From her clinical indications, the radiology technologist had suspected pleural effusion. What is the most appropriate radiographic projection for this case?
- A. PA-Oblique
 - B. AP-Oblique
 - C. Lateral Decubitus
 - D. Dorsal Decubitus
- ANSWER: C

45. A 37-year-old male patient visited a medical radiology technologist's office with a previously taken chest radiograph for interpretation. The radiograph shows ground glass opacity with a diffused increase in radiographic density resulting in the loss of the vascular pattern. Additionally, Kerley's B lines were also seen.
What is the most likely diagnosis for this case?
- A. Hemorrhage
 - B. Pulmonary infection
 - C. Pulmonary Oedema
 - D. Alveolar cell carcinoma
- ANSWER: C
46. A 27-year-old male patient was brought to an emergency department after a motor bicycle accident. After a thorough physical examination, he was sent to a radiology unit for chest x-ray. On the image, the technologist saw that the edge of the right lung is delineated by a sharply demarcated thin white line beyond which no pulmonary markings are identified, while the area around the apex of the thorax appears darker.
What is the most likely diagnosis in this case?
- A. Collapsed lung
 - B. Pneumothorax
 - C. Pleural effusion
 - D. Pneumoperitoneum
- ANSWER: B
47. A radiology technologist was evaluating a 23-year-old patient's chest x-ray, which was taken in supine position. The radiograph shows a collection of air in the left costophrenic angle, reflecting pneumothorax that is anterior and lateral.
What radiological sign indicated the finding?
- A. Meniscus
 - B. Bats wings
 - C. Deep Sulcus
 - D. Westermarck's or
- ANSWER: C
48. A 37-year-old male patient with complaints of fever and dyspnea was brought to a radiology unit for chest x-ray. On his radiograph, mass effect and space-occupying lesion in the right lung were observed. There is also a shift of the mediastinal structures toward the contralateral side.
What is the most likely diagnosis in this case?
- A. Pneumothorax
 - B. Total atelectasis
 - C. Mediastinal hematoma
 - D. Massive pleural effusion
- ANSWER: D
49. A 65-year-old male patient with a complaint of acute shortness of breath was brought to a radiology department for chest x-ray. The radiograph shows an increased heart size, increased pulmonary vascularity, redistribution of blood flow to the upper lobes, and Kerley's B-lines.
What is the most likely diagnosis in this case?
- A. Aortic dissection
 - B. Acute pneumonia
 - C. Pericardial effusion
 - D. Congestive heart failure
- ANSWER: D

50. A 45-year-old male patient with complaints of hypoxia, dyspnea, and non-productive cough had a PA chest x-ray done. Upon interpretation, the technologist saw three-to-ten-millimeter nodules present in the upper and mid zones in addition to hilar lymphadenopathy with eggshell calcification. What is the most likely diagnosis for this case?
- A. Silicosis
 - B. Teratoma
 - C. Pericardial cyst
 - D. Pancoasttumour
- ANSWER: A
51. A 38-year-old patient with complaints of cough, wheeze, and exertional dyspnea came to a radiology department for chest x-ray. The radiograph shows a large focal area of well-defined lucency outlined with a thin wall, being devoid of lung markings. It has occupied most of the right lung, compressing the adjacent lung parenchyma. What is the most likely diagnosis for this patient?
- A. Heamothorax
 - B. Pneumoconiosis
 - C. Pulmonary fibrosis
 - D. Emphysematous bulla
- ANSWER: D
52. A 60-year-old female patient visited the radiographic unit for routine mammographic follow-up. A radiology technologist starts to position her for mammographic examination. What are the basic projections in this case?
- A. Lateral and magnified view
 - B. Axillary tail and Mediolateral
 - C. Lateromedial Oblique and Craniocaudal
 - D. Mediolateral oblique and Craniocaudal
- ANSWER: D
53. A radiology technologist was evaluating a mammographic image of a 36-year-old woman who works in the modeling industry. The image demonstrates a very dense homogeneous opacity close to the chest wall, anterior to the pectoralis muscle. What is the most likely explanation for the dense opacity in this case?
- A. Breast implant
 - B. Cooper's ligament
 - C. Fibro-glandular tissue
 - D. Artifact caused by a bra
- ANSWER: A
54. A 40-year-old male patient with a medical history of pyelonephritis came to a radiology unit for IVU procedure. The radiology technologist on duty received the request paper of the patient and ordered for preparation before the procedure. On the next day, the patient arrived to the procedure room and the technologist continued to prepare the contrast agent for the procedure. What is the initial precaution to be taken in this case?
- A. Discuss the procedure with the patient
 - B. Obtain informed consent from the patient
 - C. Evaluate the patient's vital signs for abnormalities
 - D. Check whether the patient has other medical problems
- ANSWER: A

55. A radiology technologist started to perform IVU procedure. The patient is positioned, the contrast agent is then administered, and a preliminary image is taken. Immediate film was taken 10-14 seconds after contrast injection. Before the next step, the technologist saw the image in order to make a proper decision.

What is the most likely explanation in this case?

- A. Check the presence or absence of obstruction
- B. Evaluate the renal outline and possible masses
- C. Check that an adequate amount of contrast media is administered
- D. Evaluate adequate distension of the pelvicalyceal system

ANSWER: B

56. A 66-year-old male patient, with a complaint of difficulty while urinating and mild hematuria, came to a radiology department with IVU request. On the day of the examination, the radiology technologist took a scout image to perform the urogram procedure.

What is the most appropriate parameter that can be evaluated from this image?

- A. Pelvicalyceal system
- B. Anatomic structures of ureters
- C. Patient positioning & Centering
- D. To know the projection is AP or PA

ANSWER: C

57. A radiology technologist working at a hospital mentored the third-year students of medical radiology technology on special radiographic procedures. One of the students asked about the indications for the hysterosalpingography procedure, and the technologist explained accordingly.

What is the most common pathological indication in this case?

- A. Infertility
- B. Uterine mass
- C. Recurrent miscarriages
- D. Abnormal uterine bleeding

ANSWER: A

58. A radiology technologist performed cystourethrography of a male patient is taking spot images during micturition and recording reflux. After evaluating the spot images, the radiologist asked the technologist to expose well to visualize the right lower ureter for further evaluation.

What is the most appropriate positioning in this case?

- A. Left anterior oblique
- B. Left posterior oblique
- C. Right posterior oblique
- D. Right anterior oblique

ANSWER: D

59. A 45-year-old male patient was in a coma for one month. He came back to a hospital after being awake with a complaint of difficulty swallowing. The physician requested barium swallow study. The radiology technologist noticed narrowing and flaccid appearance of the esophagus during the study.

What is the most likely diagnosis for this patient?

- A. Dysphagia
- B. Diverticula
- C. Esophageal varices
- D. Reflux esophagitis

ANSWER: A

60. A patient, who was presented with malabsorption problem and dilated intestine, is sent to a radiology department for a small bowel follow-through examination. The radiology technologist started the procedure.
What is the most appropriate procedure technique to be applied?
- A. Inject barium through the catheter
 - B. Ending the examination after barium meal is performed
 - C. Obtaining radiographs at the end of the procedure
 - D. The patient ingests 200-300 mL of barium sulfate
- ANSWER: D
61. A 57-year-old male patient with severe weight loss and dyspepsia came to a radiology department for barium meal examination. The radiology technologist is ready to position the patient to view the antrum and the greater curvature of the stomach.
What is the most appropriate positioning for this patient?
- A. Prone
 - B. Oblique
 - C. Left anterior oblique
 - D. Right Anterior Oblique
- ANSWER: D
62. A 48-year-old male patient came to a hospital with a history of gastrointestinal hemorrhage. The physician ordered barium meal examination and sent him to a radiology department. A radiology technologist placed the patient in the left anterior oblique position and noticed portions of the stomach on the image.
What anatomic structures should be identified in this case?
- A. Body
 - B. Cardia
 - C. Pylorus
 - D. Lesser curve
- ANSWER: D
63. A 22-year-old female patient with a history of amenorrhea for the last one month came to an ultrasound unit for abdomino-pelvic ultrasound examination. A radiology technologist instructed the patient on preparation before the transabdominal ultrasound procedure.
What is the appropriate patient preparation in this case?
- A. Take contrast
 - B. Empty the bladder
 - C. Be in a fasting state
 - D. Distend the bladder
- ANSWER: D
64. During abdominal ultrasound examination, a radiology technologist wanted to ensure that the correct probe orientation was applied for a longitudinal scan. Then, it was confirmed that the proper orientation of the probe was applied by identifying the superior anatomical organs on the left side of the ultrasound screen.
What is the appropriate location of the probe marker in this case?
- A. Towards the patient's left
 - B. Towards the patient's feet
 - C. Towards the patient's right
 - D. Towards the patient's head
- ANSWER: D

65. A 20-year-old male patient came to an ultrasound unit for general medical checkup. During abdominal ultrasound scanning, a radiology technologist noted isoechoic, tongue-like elongation of the right lobe of the liver extending inferiorly and lying anterior to the right kidney.

What is the most likely explanation in this case?

- A. Caudate lobe
- B. Riedel's lobe
- C. Hepatomegaly
- D. Left lobe variation

ANSWER: B

66. A 29-year-old male patient came to an ultrasound unit for medical checkup. Abdominal ultrasound showed the liver parenchyma echotexture is slightly brighter than the adjacent renal cortex, otherwise fine and uniform echo pattern.

What is the most likely diagnosis in this case?

- A. Hepatitis
- B. Fatty Liver
- C. Fatty sparing
- D. Fatty Cirrhosis

ANSWER: B

67. A 40-year-old female patient came to an ultrasound unit with complaints of RUQ Pain, fever, and jaundice. Abdominal ultrasound showed a cystic lesion containing multiple peripheral daughter cysts and central heterogeneous content in the left lobe of the liver.

What is the most likely diagnosis in this case?

- A. Simple cyst
- B. Hydatid cyst
- C. Pyogenic abscess
- D. Hemorrhagic Cyst

ANSWER: B

68. A 25-year-old female patient came to an ultrasound unit for medical checkup. Abdominal ultrasound showed diffuse inhomogeneous and coarse parenchymal echotexture with numerous small, hypoechoic and hyperechoic foci uniformly distributed throughout the liver.

What is the most likely diagnosis in this case?

- A. Caroli disease
- B. Biliary Hamartoma
- C. Hepatic microabscesses
- D. Multiple simple hepatic cysts

ANSWER: B

69. A 23-year-old female patient came to an ultrasound unit for medical checkup. Abdominal ultrasound showed mobile, gravity-dependent, without acoustic shadowing, amorphous, and mid-level echoes within the gallbladder.

What is the most likely diagnosis in this case?

- A. Adenomyomatosis
- B. Cholelithiasis
- C. Sludge
- D. Polyp

ANSWER: C

70. A 63-year-old female patient came to an ultrasound unit with a history of intermittent RUQ pain, anorexia, and weight loss. Ultrasound showed a thickened gallbladder wall containing multiple cystic intramural cavities (Rokitansky-Aschoff sinuses). There were also tiny echogenic intramural foci in the GB wall producing “comet tail” artifacts.
What is the most likely diagnosis for this case?
A. Adenomyomatosis
B. Chronic cholecystitis
C. Adenomatous polyp
D. Emphysematous cholecystitis
ANSWER: A
71. A 63-year-old female patient came to an ultrasound unit with a history of intermittent RUQ pain, anorexia, and weight loss. Ultrasound showed multiple echogenic foci with posterior shadowing, diffuse wall thickening with hypoechoic intramural nodules, and continuous mucosal line.
What is the most likely diagnosis for this patient?
A. Acute Cholecystitis
B. Chronic Cholecystitis
C. Gangrenous cholecystitis
D. Xanthogranulomatous Cholecystitis
ANSWER: D
72. A 47-year-old female patient came to an ultrasound unit with a history of tenderness and palpable mass in LUQ. Abdominal ultrasound showed a well-defined anechoic lesion with posterior acoustic enhancement.
What is the most likely diagnosis in this case?
A. Splenosis
B. Lymphoma
C. Splenic Cyst
D. Lymphangioma
ANSWER: C
73. A 33-year-old male patient came to an ultrasound unit for medical checkup. Ultrasound showed a well-circumscribed, smooth-walled unilocular anechoic mass with posterior acoustic enhancement in the pancreatic tail.
What is the most likely diagnosis in this case?
A. Chronic Pancreatitis
B. Cystic islet cell tumor
C. Pancreatic Pseudocyst
D. Autoimmune pancreatitis
ANSWER: C
74. A 20-year-old male patient came to an ultrasound unit for medical checkup. Abdominal ultrasound showed external bulge on the renal contour. The cortex remains constant in thickness, with hypoechoic pseudotumor composed of normal renal tissue.
What is the most likely explanation for this finding?
A. Renal Ectopia
B. Dromedary hump
C. Column of Bertin
D. Renal Junction Line
ANSWER: B

75. A 35-year-old male patient came to an ultrasound unit with a complaint of acute flank pain. During abdominal ultrasound scanning, a radiology technologist identified echogenic foci with sharp posterior acoustic shadowing within the mid-right kidney.
What is the most likely diagnosis in this case?
- A. Nephrolithiasis
 - B. Nephrocalcinosis
 - C. Papillary necrosis
 - D. Emphysematous pyelonephritis
- ANSWER: A
76. A 66-year-old male patient came to an ultrasound unit for medical checkup. Abdominal ultrasound showed a well-defined, thin-walled, round anechoic lesion without internal echoes and posterior acoustic enhancement.
What is the most likely diagnosis for this patient?
- A. Simple Renal Cyst
 - B. Complex renal cyst
 - C. Prominent pyramids
 - D. Perinephric collections
- ANSWER: A
77. A 24-year-old female patient came to an ultrasound unit with a history of recurrent chronic infection with insufficiency. Abdominal ultrasound illustrates reduced right kidney in size with increased echogenicity. The left kidney has normal sonographic features.
What is the most likely diagnosis in this case?
- A. Acute Pyelonephritis
 - B. Chronic Pyelonephritis
 - C. Acute Renal Infarction
 - D. Acute Tubular Necrosis
- ANSWER: B
78. A 32-year-old male patient came to an ultrasound unit with a history of abdominal discomfort, distension, and weight gain due to end-stage liver disease. Abdominal ultrasound showed free fluid collection surrounding the bowel without mass effect.
What is the most likely diagnosis in this case?
- A. Ascites
 - B. Effusion
 - C. Hemoperitoneum
 - D. Pseudomyxoma peritonei
- ANSWER: A
79. A 45-year-old male patient came to an ultrasound unit with a history of car accident. During ultrasound scanning, a radiology technologist identified a non-spherical shaped, complex mass that is solid with some central liquefied regions on the parenchyma of the spleen.
What is the most likely diagnosis in this case?
- A. Cyst
 - B. Abscess
 - C. Hematoma
 - D. Lymphoma
- ANSWER: C

80. 45-year-old women with suspected endometrial disorders and a family history of ovarian cancer was presented to an ultrasound room for pelvic ultrasound examination. A radiology technologist identified an echogenic structure within the endometrium using trans-abdominal transducer. The technologist informed the patient about the procedure and rescanned her using transvaginal transducer.

What is the most appropriate reason for rescanning in this case?

- A. To examine with an empty bladder
- B. To better visualize the surrounding structures
- C. For greater detail of the internal characterization of the mass
- D. For visualization of the mass in a larger field of view using high frequencies

ANSWER: C

81. A senior radiology technologist performed a pelvic ultrasound examination, using transvaginal sagittal scanning with anterior angulation to better visualize the fundus of a normal anteflexed uterus. Then, the technologist continued to record in the coronal plane to obtain a coronal section of the uterus.

What is the most appropriate next step in this case?

- A. Gently rotate the transducer 30 degrees
- B. Gently rotate the transducer 60 degrees
- C. Gently rotate the transducer 90 degrees
- D. Gently rotate the transducer 180 degrees

ANSWER: C

82. A 38-year-old female patient with a history of infertility came to a radiology room for an ultrasound of the pelvic organs. Longitudinal trans-abdominal ultrasound demonstrated well distended urinary bladder and the body as well as fundus of the uterus tipped posteriorly with linearly oriented cervix and vagina.

What is the most appropriate orientation of the uterus in this case?

- A. Anteflexed
- B. Anteverted
- C. Retroflexed
- D. Retroverted

ANSWER: D

83. A 65-year-old female patient with a complaint of pelvic pain came to a radiology room for ultrasound examination. The ultrasound finding showed oval shaped, thin-walled cystic structure on the posterior wall of the urinary bladder, and the radiology technologist observed a change in size of structure on real-time scanning.

What is the most likely diagnosis in this case?

- A. Ureterocele
- B. Urachal cyst
- C. Cystitis cystica
- D. Bladder diverticula

ANSWER: A

84. A 50-year-old male patient with a chief complaint of pain during urination and urine discoloration came to an ultrasound room for pelvic ultrasound examination. Ultrasound was performed after proper preparation. The results revealed a mobile, echogenic focus with strong posterior shadowing within the urinary bladder.

What is the most likely diagnosis in this case?

- A. Blood clot
- B. Bladder sludge
- C. Bladder calculi
- D. Bladder carcinoma

ANSWER: C

85. A 75-year-old male patient came to the ultrasound room with complaints of urination and urgency for ultrasound screening of the prostate. After preparing the patient, a radiology technologist identified a focal hypoechoic and solid nodule in the left peripheral zone. A focus of calcification was also seen in the central gland on the transverse scan of the prostate. What is the most likely diagnosis in this case?
- A. Prostate cyst
 - B. Prostate cancer
 - C. Benign Prostatic Hyperplasia
 - D. Normal prostate ultrasound finding
- ANSWER: B
86. A 42-year-old female patient, with complaints of abnormally heavy, prolonged menses and pelvic pain, especially around menstruation, presented to a radiology room for a gynecology ultrasound. Ultrasound scans of the uterus revealed an enlarged uterine size with diffuse heterogeneous echotexture, and multiple myometrial cysts were also seen. What is the most likely diagnosis in this case?
- A. Uterine fibroids
 - B. Uterine sarcomas
 - C. Uterine carcinomas
 - D. Uterine adenomyosis
- ANSWER: D
87. A 47-year-old female patient with a complaint of vaginal bleeding presented to a radiology room for ultrasound examination. Ultrasound findings showed a focal, echogenic, and elongated lesion disrupting the normal echotexture of the endometrium. What is the most likely diagnosis in this case?
- A. Endometrial hyperplasia
 - B. Endometrial carcinoma
 - C. Endometrial atrophy
 - D. Endometrial polyp
- ANSWER: D
88. A 36-year-old female patient with a complaint of pelvic pain presented to an ultrasound room for a gynecology ultrasound examination. The ultrasound examination revealed cystic lesion with thick wall, low-level internal echoes, a smooth posterior wall, and posterior acoustic enhancement in the left adnexa. What is the most likely diagnosis in this case?
- A. Surface epithelial inclusion cyst
 - B. Corpus luteum cyst
 - C. Hemorrhagic cyst
 - D. Follicular cyst
- ANSWER: B
89. A 25-year-old female patient, with acute onset of severe unilateral lower abdominal pain of one hour duration, came to an emergency ultrasound unit for examination. Ultrasound examination revealed a large, rounded, and heterogeneous echo-texture mass on the right adnexa measuring 6.0cm, with asymmetric enlargement of the right ovary on the side of pain compared with the contralateral side. There is also free fluid in the cul-de-sac. What is the most likely diagnosis in this case?
- A. Ovarian torsion
 - B. Ruptured ovarian cyst
 - C. Tubo-ovarian abscess
 - D. Ovarian Ectopic pregnancy
- ANSWER: A

90. A 27-year-old female patient, with a complaint of amenorrhea of two months duration, came to a radiology room for ANC ultrasound visit. The transvaginal ultrasound finding revealed a small ring within the gestational sac with highly echogenic borders at five weeks of gestational age.

What is the most likely structure visualized in this case?

- A. Embryo
- B. Yolk sac
- C. Amniotic sac
- D. Chorionic membrane

ANSWER: B

91. A 27-year-old female patient came to a radiology unit with a chief complaint of amenorrhea of one month duration. Ultrasound findings revealed an intrauterine pregnancy and well well-defined gestational sac without visible embryo.

What is the most likely parameter used in this case?

- A. MSD
- B. CRL
- C. BPD
- D. FL

ANSWER: A

92. A 23-year-old pregnant woman at the 8th week of gestation presented to a radiology room with abdominal cramps and spotting of five days duration. Trans-abdominal ultrasound scan showed a gestational sac in the lower uterine segment extending into the cervix without cardiac activity. What is the most likely diagnosis in this case?

- A. Abortion
- B. Ectopic pregnancy
- C. Embryonic demise
- D. Anembryonic pregnancy

ANSWER: A

93. A 24-year-old pregnant women came to a radiology room for obstetric ultrasound examination with a chief complaint of vaginal bleeding. An ultrasound finding showed enlargement of the uterus with mixed echo texture, including an echoic fluid collection.

What is the most likely diagnosis in this case?

- A. Hydatidiform mole
- B. Choriocarcinoma
- C. Uterine fibroid
- D. Abortion

ANSWER: B

94. A 21-year-old female patient presented to an ultrasound room for ultrasound examination with chief complaints of lower abdominal pain and vaginal bleeding. An ultrasound finding revealed an enlarged yolk sac size greater than six millimeters in diameter between five and ten weeks of gestational age.

What is the most likely diagnosis in this case?

- A. Normal early pregnancy
- B. Embryonic demise
- C. Ectopic pregnancy
- D. Abortion

ANSWER: B

95. A 28-year-old pregnant woman was presented to a radiology unit for ANC ultrasound visit. A medical radiology technologist observed that there is a singleton intrauterine pregnancy with gestational age of 17, 18, and 19 weeks by BPD, FL, and AC measurements, respectively.

What is the most accurate parameter used in this case?

- A. BPD
- B. FL
- C. AC
- D. HC

ANSWER: A

96. A 23-year-old pregnant woman was presented to a radiology unit for ANC ultrasound follow-up. The obstetrician requested obstetric ultrasound examination for assessment of placental location, amniotic fluid volume, and congenital anomalies of the fetus.

What is the most appropriate scanning time in this case?

- A. First-trimester pregnancy
- B. Second-trimester pregnancy
- C. Third-trimester pregnancy
- D. Term pregnancy

ANSWER: B

97. A 36-year-old pregnant woman at 30 weeks' of gestation was brought to a radiology room for obstetric ultrasound examination with a history of vaginal bleeding. She has previous history of caesarean delivery. An ultrasound finding revealed loss of the hypoechoic border between the placenta and the myometrium with protrusion of the placenta into the myometrium.

What is the most likely diagnosis in this case?

- A. Placenta percreta
- B. Placenta accreta
- C. Placenta increta
- D. Placenta previa

ANSWER: C

98. A 28-year-old pregnant woman at 31th week of gestational age came to a radiology room with chief a complaint of vaginal bleeding. An ultrasound finding showed that the placenta tissue is located at the edge of the internal cervical os.

What is the most likely diagnosis in this case?

- A. Marginal placenta previa
- B. Low-lying placenta previa
- C. Complete placenta previa
- D. Partial placenta previa

ANSWER: A

99. A 27-year-old pregnant woman at 29th week of gestation came to a radiology room for ANC ultrasound examination with a chief complaint of vaginal bleeding. An ultrasound finding revealed that a segment of the cervical os was obstructed by overlying placenta.

What is the most likely diagnosis in this case?

- A. Low-lying placenta previa
- B. Complete placenta previa
- C. Marginal placenta previa
- D. Partial placenta previa

ANSWER: D

100. A 32-year-old pregnant woman at 30th week of gestational age came to an ultrasound room for ANC ultrasound visit with chief complaints of lower abdominal pain and decreased fetal movement. The obstetrician requests biophysical profile studies to assess the fetal well-being. An ultrasound finding revealed that 8/10 test scores with the abnormal amniotic fluid volume.

What is the most likely diagnosis in this case?

- A. Chronic asphyxia
- B. Normal finding
- C. Acute asphyxia
- D. IUGR

ANSWER: A

101. A 22-year-old pregnant woman at 35th weeks of gestation came to a radiology unit for ANC ultrasound visit. An ultrasound finding revealed estimated amniotic fluid volume of the largest vertical pool measures of 1.8cm, and the AFI was 3.0cm.

What is the most likely diagnosis in this case?

- A. Oligohydramnios
- B. Polyhydramnios
- C. Adequate AFV
- D. Anhydramnios

ANSWER: A

102. A 32-year-old pregnant woman at 27th weeks of gestation came to a radiology room for ANC visit. An ultrasound finding showed a unilateral enlarged kidney containing multiple anechoic lesions of varying size, with a reduced amniotic fluid volume with decreased fetal heartbeat.

What is the most likely cause for this finding?

- A. Rhesus disease
- B. Duodenal atresia
- C. Diaphragmatic hernia
- D. Polycystic kidney disease

ANSWER: D

103. A 34-year-old pregnant woman at 18th weeks of gestation came to a radiology room for ANC ultrasound visit. An ultrasound finding revealed a twin intrauterine pregnancy with placental "T" sign appearance.

What is the most likely diagnosis in this case?

- A. Monochorionic-monoamniotic twin pregnancy
- B. Monochorionic-diamniotic twin pregnancy
- C. Dichorionic-monoamniotic twin pregnancy
- D. Dichorionic-diamniotic twin pregnancy

ANSWER: B

104. A 23-year-old pregnant woman at 28th weeks of gestation came to a radiology room for ANC ultrasound visit. An ultrasound finding showed twin intrauterine pregnancy with positive fetal heartbeats and one fetus with intrauterine growth restriction, an empty bladder, and unhydramnios, and the other fetus was polyhydramnios associated with a stuck twin appearance.

What is the most likely diagnosis in this case?

- A. Twin-to-twin embolization syndrome
- B. Twin-to-twin transfusion syndrome
- C. A cardiac parabolic twin
- D. Conjoined twins

ANSWER: B

105. A 34-year-old pregnant woman at 26th weeks of gestation came to a radiology room for ANC ultrasound visit. An ultrasound finding showed a singleton intrauterine pregnancy with a positive fetal heartbeat and transverse section of the fetal head demonstrating scalloping of the frontal bones, described as the 'lemon-shaped' head.

What is the most likely diagnosis in this case?

- A. Anencephaly
- B. Encephalocele
- C. Hydrocephalus
- D. Chiari II malformation

ANSWER: D

106. A 23-year-old pregnant woman at 17th week of gestation came to an ultrasound room for ANC ultrasound visit. An ultrasound finding revealed a fluid-filled cranium with the presence of falx and thalamus, basal ganglia, and an intact brain stem. However, no cerebral tissue is seen.

What is the most likely diagnosis in this case?

- A. Hydrocephalus
- B. Hydranencephaly
- C. Subdural hygroma
- D. Holoprosencephaly

ANSWER: B

107. A 23-year-old pregnant woman at 19th weeks of gestation came to an ultrasound room for ANC ultrasound visit. Transverse section of the fetal abdomen demonstrated an anterior abdominal wall defect with protrusion of stomach and liver contents that were covered by peritoneum and amnion. In addition, the umbilical cord was directly inserted into the apex of the defect.

What is the most likely diagnosis in this case?

- A. Gastroschisis
- B. Omphalocele
- C. Umbilical hernia
- D. Ectopic vesicae

ANSWER: B

108. A 75-year-old male patient was appointed for a triphasic abdominal CT scan. The patient was well-prepared according to the pre-procedural patient preparation protocols for contrast-enhanced examinations. The radiology technologist then started to prepare the contrast for injection and decided to warm the contrast up to the level of body temperature before injection.

What is the most likely reason for warming the contrast?

- A. To increase viscosity
- B. To decrease viscosity
- C. To increase osmolality
- D. To decrease osmolality

ANSWER: B

109. A radiology technologist wants to perform a head CT scan of a patient with a history of a posterior fossa lesion. From his previous experience, he has faced common inherent streak artifacts specifically on the posterior fossa due to the difference in beam attenuation ability of very dense bone and much less dense tissue of the brain.

What is the most appropriate technique to reduce the artifact in this case?

- A. Increasing slice thickness and KVP
- B. Decreasing slice thickness and KVP
- C. Increasing slice thickness and decreasing KVP
- D. Decreasing slice thickness and increasing KVP

ANSWER: D

110. A pediatric patient with a history of hydrocephalus came to a radiology department for a head CT scan. The radiology technologist performing the scan proceeded to the image acquisition step.

What is the most appropriate scan plane to be employed in this case?

- A. Sagittal
- B. Coronal
- C. Oblique
- D. Transverse

ANSWER: D

111. After performing a head CT scan, a radiology technologist noticed a hyperdense lesion that was distributed along the inner table of the left frontoparietal skull bone. The lesion is easily depictable with a concave medial border that extends across the cranial sutures.

What is the most likely diagnosis for this case?

- A. Epidural haematoma
- B. Intra axial haematoma
- C. Sub dural haematoma
- D. Sub arachnoid haematoma

ANSWER: C

112. A radiology technologist was performing a CT scan of the neck in a single-slice CT scanner. He selected axial scan mode from the available protocol for neck soft tissue and used narrow collimation to the area of interest for better image resolution.

What is the most likely parameter affected by the collimation in this case?

- A. Pitch
- B. Table feed
- C. Inter-slice gap
- D. Slice thickness

ANSWER: D

113. After completing a PNS CT scan with contrast, the radiology technologist was evaluating the images. On the coronal section, he noticed a soft tissue mass on the anterior ethmoidal air cells with peripheral ring-like enhancement on the post-contrast sequence.

What is the most likely finding in this case?

- A. Polyp
- B. Mucocoele
- C. Paget disease
- D. Fibrous dysplasia

ANSWER: B

114. An adult patient, who sustained a road traffic accident two hours ago, came to a referral hospital. The physician ordered a lumbar spine CT scan to rule out lumbar vertebral fracture, but the radiology technologist noticed that the patient is claustrophobic of the gantry system. The radiology technologist examined by reassuring and adjusting the patient's positioning. What is the most likely patient position applied in this case?
- A. Prone feet first
 - B. Supine head first
 - C. Prone head first
 - D. Spine feet first
- ANSWER: D
115. A 70-year-old patient with a complaint of pain over the thoracic and lumbar spine came to a referral hospital. The physician examined the patient and ordered thoraco-lumbar CT myelography. The radiology technologist accepted the request, positioned the patient on the scan table, and started adjusting the protocol to suit the requested examination. What is the most appropriate slice thickness to be suggested in this case?
- A. Two mm
 - B. Three mm
 - C. One mm
 - D. Five mm
- ANSWER: D
116. A 65-year-old patient with a complaint of chronic back pain presented to a radiology department for a spine CT scan to rule out degenerative disk disease. The radiology technologist is going to examine the administration of sufficient dilute contrast medium. What is the most appropriate delay time in this case?
- A. 1 to 3 hours
 - B. 2 to 4 hours
 - C. 3 to 5 hours
 - D. 4 to 6 hours
- ANSWER: A
117. An adult patient with a history of chronic cough and fast breathing presented to a radiology department for a chest CT scan to rule out interstitial lung disease. What is the most appropriate scanning technique required for this case?
- A. Routine chest CT
 - B. HRCT with thin slice
 - C. Chest CT Angiography
 - D. Low-dose Volumetric CT
- ANSWER: B
118. A female medical radiology technologist, who works in a referral hospital in a CT scan unit, received a patient with a chest CT scan request. The technologist started performing a CT scan examination. After viewing the image, she decided not to administer contrast media. What is the most likely indication for this case?
- A. Mediastinal abnormalities
 - B. Pulmonary embolism
 - C. Pleural mass
 - D. Metastasis
- ANSWER: D

119. A radiology technologist performed a contrast bolus scan of the chest for a 20-year-old patient. The technologist took one cross-sectional image at the level of the aortic arch to demonstrate different anatomical structures. On the image, a triangular and homogenous soft tissue structure was visualized in the anterior mediastinum anterior to the aortic arch.

What is the most likely anatomy demonstrated in this case?

- A. Trachea
- B. Esophagus
- C. Thymus
- D. Thyroid

ANSWER: C

120. An adult patient with complaints of chest pain, cough, and fast breathing came to a radiology department for a chest CT scan. The radiology technologist performed the examination by modifying the protocol to scan the sequence in one breath hold.

What is the most likely pitch factor applied to this case?

- A. 0.5
- B. 0.8
- C. 1.0
- D. 1.5

ANSWER: D

121. An adult patient with a history of abdominal distension came to a radiology department for abdominopelvic CT scan. The radiology technologist examined the patient by informing him to hold his breath during the data acquisition.

What is the most likely reason for the breathing instruction in this case?

- A. Decrease scan time
- B. Remove streaking artefact
- C. Increase the scan field of view
- D. Reduce anatomic structure blurring

ANSWER: D

122. An adult patient with a complaint of right upper quadrant pain and abdominal swelling of two years duration presented to a radiology department for abdominal CT scan to rule out hyper-vascular tumors of the liver.

What is the most appropriate scanning phase for this case?

- A. Portal venous phase
- B. Angiographic phase
- C. Late arterial phase
- D. Delayed phase

ANSWER: C

123. A 60-year-old male patient with a complaint of urinary incontinence of two weeks duration came to a radiology department for a pelvis CT scan. The radiology technologist performed the examination and evaluated cross section axial image and demonstrated a homogenous soft tissue density anatomical structure just anterior to the rectum on the image.

What is the most likely structure demonstrated in this case?

- A. Symphysis pubis
- B. Urinary bladder
- C. Psoas muscle
- D. Prostate

ANSWER: D

124. An adult patient with a complaint of bilateral flank pain of six-month duration came to a radiology department for CT urography. After the scanogram, a radiology technologist performed the pre-contrast scan sequence.

What is the most appropriate reason for performing this sequence?

- A. Plan for the next series
- B. Reduce the radiation dose
- C. Identify the normal anatomy
- D. Search for calcification and stone

ANSWER: D

125. A basketball player who sustained trauma to the wrist came to a radiology department for a right wrist CT scan to rule out scaphoid fracture. The radiology technologist positioned the patient on the scanning table and started to modify the scan parameters for the coronal and sagittal planes.

What is the most appropriate slice thickness for data acquisition in this case?

- A. 0.5 mm
- B. 2 mm
- C. 3.5 mm
- D. 4 mm

ANSWER: A

126. An adult patient who sustained trauma to the left ankle came to a radiology department for left ankle CT scan. The radiology technologist noticed that the patient cannot extend the knee during positioning the patient on the scan table.

What is the most appropriate approach for this case?

- A. Performing with MRI
- B. Using gantry angulation
- C. Gently extending the knee
- D. Cancelling the examination

ANSWER: B

127. A radiology technologist performed a CT scan of the right ankle and foot for an adult male patient who sustained trauma to the right ankle joint. On the axial images, the radiologic technologist observed fracture of the talar neck and wanted to see the alignment of the tarsal and ankle joints. Then, the radiology technologist started to perform a post-processing reconstruction.

What is the most appropriate reformat recommended for this case?

- A. MIP and MinIP
- B. Volume rendering
- C. Sagittal and coronal
- D. Surface shaded display

ANSWER: C

128. A 60-year-old female patient came to a radiology department with a request for thoracic aorta angiography. The radiology technologist positioned the patient and started to perform the examination.

What is the most appropriate scan coverage for this case?

- A. Carina to hepatic artery
- B. Base of neck to celiac artery
- C. Level of sternal notch to renal artery
- D. Aortic arch to inferior mesenteric artery

ANSWER: B

129. A 40-year-old female patient came to a radiology department with a request for pulmonary angiography CT. The radiology technologist prepared and positioned the patient on the scan table. Then, the technologist wanted to scan the patient by decreasing the respiratory motion artefact.

What is the most appropriate scanning direction in this case?

- A. Cranial-to-caudal
- B. Caudal-to-cranial
- C. Right-to-left
- D. Left-to-right

ANSWER: B

130. A 24-year-old female patient came to an MRI center for imaging of the head. As per the instruction given by a radiology technologist, the patient removed all her clothes and metal, including her jewelry, and then wore the MRI gown.

What is the most appropriate next step to be taken before taking the patient into the MRI room?

- A. Checking for helium leakage
- B. Assessing the magnet's temperature
- C. Choosing the correct coil for the exam
- D. Scanning the patient with a handheld metal detector

ANSWER: D

131. A 41-year-old male patient came to a hospital with a severe headache for the last two years. The clinician orders MRI for brain tumor staging. The patient has impaired renal function and has a previous drug reaction to gadolinium. The technologist performing the scan has taken this into consideration.

What is the most appropriate measure to be taken in this case?

- A. Perform the scan with contrast
- B. Use another type of contrast agent
- C. Perform the scan without contrast
- D. Scan with only half dose of contrast

ANSWER: C

132. A 35-year-old female patient with severe headache, visual disturbance, and unexpected breast milk production presented to a radiology department. The clinician orders an MR brain scan with the indication to rule out an intracranial space-occupying lesion. Based on the clinical signs, the radiology technologist focused the planning on the inferior edge of the sphenoid sinus to the superior portion of the lateral ventricles on the sagittal plane.

What is the most likely anatomic structure that the technologist included in the planning?

- A. Orbit
- B. Pituitary
- C. Internal ear
- D. Corpus callosum

ANSWER: B

133. A 45-year-old man was presented to a hospital with confusion and symptoms of severe sepsis. Shortly after admission, he was brought to the MR unit for a brain scan, and the technologist positioned him on the examination couch. While scanning, the patient was uncooperative and kept disrupting the process.

What is the most appropriate measure the technologist should apply for this case?

- A. Give the patient an alarm bell
- B. Give the patient headphones
- C. Use immobilization pads and straps
- D. Apply high-performance gradients for EPI

ANSWER: C

134. A 55-year-old male patient came to an MRI center with a history of multiple sclerosis. A brain MR scan from a previous report showed a gray-white matter lesion and CSF abnormality. A radiology technologist planned to perform a scan using protocols sufficient to demonstrate this condition.

What is the most appropriate sequence to be used in this case?

- A. FLAIR
- B. STIR
- C. DWI
- D. EPI

ANSWER: A

135. A patient with a persistent headache came to a radiology department for a scan. Non-contrast T1W MR of the brain showed a hypo-intense lesion over the frontal lobe, and the T2W image shows a heterogeneous appearance circumscribed by a thin capsule with signal intensity. Following contrast administration, the lesion was well enhanced uniformly; distortion of the adjacent brain parenchyma with compression of the left lateral ventricle and mild shift of the midline structures to the right is noted.

What is the most likely lesion visualized on the image in this case?

- A. Meningioma
- B. Petrous bone cyst
- C. Subdural haematoma
- D. Arachnoid granulation

ANSWER: A

136. A brain MRI image of a patient with headache was scanned by a radiology technologist. The T2-weighted image showed uneven contrast with loss of signal intensity in one part of the image. The technologist has noted that this is a shading artifact.

What is the most appropriate corrective measure to be taken in this case?

- A. Reduce scan time
- B. Increase inter-slice gap
- C. Loading the coil correctly
- D. Use a fat suppression sequence

ANSWER: C

137. A 42-year-old female patient came with shoulder pain and limited range of movement. The physician suspected a rotator cuff tear and sent her to MRI center for a shoulder MR scan. The technologist wants to position the patient using the correct positioning technique.

What is the most appropriate technique to be applied in this case?

- A. Prone with the arms above the head
- B. Prone with the arms resting by the side
- C. Supine with the arms above the head
- D. Supine with the arms resting by the side

ANSWER: D

138. A 31-year-old patient with a complaint of ankle instability and repeated ankle sprains is presented to a hospital. The physician suspected lateral ankle ligament injury and ordered an MR scan. A radiology technologist then started to plan an axial scan.

What anatomic area should be included in this planning technique?

- A. From lateral to medial aspects of the ankle
- B. Distal Achilles tendon to the bottom of the talus
- C. From distal tibia to calcaneus and sole of the foot
- D. From the origin of the Achilles tendon to the inferior calcaneus

ANSWER: D

139. A 51-year-old patient with a complaint of bilateral hip pain was sent to MRI center for a hip joint MR scan. The radiology technologist prescribed slices from the posterior to the anterior margins of the musculature of the hip.

What is the most likely plane applied by the technologist in this scanning technique?

- A. Axial
- B. Oblique
- C. Sagittal
- D. Coronal

ANSWER: D

140. A 52-year-old male patient suspected of having a distal common bile duct stone was presented to MRI center for Magnetic Resonance Cholangiopancreatography examination. A radiology technologist wants to perform the scan after preparing the patient.

What is the most appropriate patient preparation for this procedure?

- A. Oral water contrast
- B. 8-12 hours of fasting
- C. Administer Lasix medication
- D. Physical exercise before the exam

ANSWER: B

141. A 42-year-old male patient with a history of intra-abdominal mass presented to MRI center for abdominal MR examination. A radiology technologist positioned the patient on the couch and applied the proper respiratory compensation technique.

What is the most appropriate location for the respiratory gating should be placed?

- A. Level of the lung apex
- B. Level of diaphragm
- C. Level of the umbilicus
- D. Level of pubic symphysis

ANSWER: B

142. A 35-year-old female patient with a history of pelvic pain and menstrual irregularities came to an MRI center. A pelvic MR scan was performed, and the radiology technologist identified an ovarian lesion with a hyper-intense signal on both T1 and T2-weighted images.

What is the most likely pathological structure identified in this case?

- A. Fibroid
- B. Adnexal mass
- C. Endometrioma
- D. Hemorrhagic cyst

ANSWER: D

143. A 30-year-old female patient with a history of pelvic inflammatory disease came to an MRI center for a pelvic MR scan. Her scan was performed, and the radiology technologist identified a central stripe with high signal intensity on the T2-weighted sagittal image while visualizing the anatomy of the uterus.

What is the most likely anatomical structure identified in this case?

- A. Cervix
- B. Myometrium
- C. Endometrium
- D. Perimetrium

ANSWER: C

144. A 51-year-old male patient presented to a hospital with complaints of dizziness and sudden onset of left-sided body weakness. He has a history of hypertension. Then, the patient was sent to MRI center for a brain angiography study. A radiology technologist planned to scan the axial 3D time of flight protocol from the localizer image.

What is the most appropriate planning technique for this procedure?

- A. From axial only
- B. From sagittal and axial
- C. From axial and coronal
- D. From sagittal and coronal

ANSWER: D

145. A 31-year-old female patient complains of lower back pain following trauma. The physician suspected disc herniation and sent her to MRI center for a thoracic MR scan. A radiology technologist positioned the patient and planned slices from the left to the right lateral borders of the vertebral bodies on the coronal image.

What is the most likely planning technique applied by the technologist in this case?

- A. Axial
- B. Oblique
- C. Coronal
- D. Sagittal

ANSWER: D

146. A 25-year-old male patient with a complaint of upper back pain following a sport injury is presented to a hospital. The physician suspected disc prolapse and sent him to MRI center for a thoracic spine scan. A radiology technologist positioned the patient and planned slices parallel to the disc space or perpendicular to the lesion on the sagittal image.

What planning technique was applied by the technologist in this case?

- A. Axial
- B. Oblique
- C. Sagittal
- D. Coronal

ANSWER: A

147. A 32-year-old woman had a lumbar MRI scan for a suspected case of bone marrow abnormality. The technologist performing the scan wants to select a sequence that allows demonstration of metastatic disease.

What is the most appropriate sequence to be selected in this case?

- A. T1 weighted
- B. T2 weighted
- C. Short T1 inversion recovery
- D. Diffusion-weighted imaging

ANSWER: C

148. A 53-year-old female patient with neck pain and cervical myelopathy symptoms presented to an MRI center for pre- and post-contrast cervical examination. A radiology technologist started to plan the coronal plane and selected an anatomical structure to perform cervical myelography.

What anatomical structure was used as a landmark for this procedure?

- A. CSF
- B. Discs
- C. Spinal cord
- D. Spinous process

ANSWER: C

149. A medical physicist performed an acceptance test for a newly installed CT machine. The physicist taped the film to the backing plate with the film edge parallel to the plate edge. Exposed and processed the film. Finally, the physicist measured the distance from the holes of the processed film to the midpoint of the radiation field.

What is the most likely acceptance test performed in this case?

- A. Table/Gantry Tilt
- B. Radiation Profile Width
- C. Low-Contrast Performance
- D. Scan Localization Light Accuracy

ANSWER: D

150. A radiology department purchased imaging plates for computed radiography. The quality assurance technologist performed an acceptance test for the new imaging plate. The technologist put the wire mesh on the imaging plate, exposed and processed it to identify the presence of image distortion.

What is the most likely acceptance test performed in this case?

- A. Uniformity
- B. Dark Noise
- C. Spatial Accuracy
- D. Erasure Thoroughness

ANSWER: C

151. A radiology technologist selected and used appropriate exposure factors while performing a routine examination. However, the technologist observed magnification on all images.

What is the most appropriate quality control test to be performed in this case?

- A. Radiation output consistency
- B. X-ray to light field alignment
- C. Source-to-image distance indication
- D. Automatic exposure control constancy

ANSWER: C

152. A radiology technologist exposed the imaging plate of the computed radiography using appropriate exposure factors. The technologist processed the exposed imaging plate with the CR reader and assessed the image for artifacts.

What is the most likely quality control test done in this case?

- A. Uniformity
- B. Dark Noise
- C. Spatial Accuracy
- D. Erasure Thoroughness

ANSWER: A

153. A radiology technologist performed a quality control test for computed tomography using a phantom containing uniform objects of known materials with a wide range of CT numbers. After scanning this phantom, the technologist compared the measured CT numbers with the nominal values provided by the manufacturer.

What is the most likely quality control test employed in this case?

- A. CT number linearity
- B. CT number accuracy
- C. CT number uniformity
- D. CT number calibration

ANSWER: A

154. A radiology technologist, who works in a magnetic resonance imaging unit, started his daily work by asking about the patient's history and preparation of the patient for scanning in the recommended MRI zone.

What is the most likely location of the radiology technologist in this case?

- A. MR zone I
- B. MR zone II
- C. MR zone III
- D. MR zone IV

ANSWER: B

155. A female radiology technologist who works in the magnetic resonance unit has a known pregnancy in the first trimester.

What is the appropriate safety decision to be applied in this case?

- A. Immediately stop her work
- B. Continue her work until the third trimester
- C. Leave from work in the first and early second trimester
- D. Can work in the MRI room throughout her pregnancy

ANSWER: D

156. A patient came to the X-ray unit for an X-ray examination. A technologist selected low exposure factors to reduce the patient's dose while producing high image quality.

What is the most likely radiation protection principle applied in this case?

- A. Distance
- B. Limitation
- C. Justification
- D. Optimization

ANSWER: D

157. A head of a health center has developed a plan for expanded immunization program that aims at vaccinating all children in the community. The head has arranged training for vaccinators in collaboration with other concerned bodies. Activities are sequentially arranged, and a multidisciplinary health team is involved for the general effectiveness of the program.

What is the most likely management principle applied in this case?

- A. Division of Labor
- B. Functional Structure
- C. Convergence of Work
- D. Management by Exception

ANSWER: C

158. A radiology technologist, who is appointed as new head of the radiology department, communicated with all staff of the department on the effectiveness of each department in executing its activities and gave recognition to best best-performing radiographers. In addition to this, the head attempted to build cooperation among the staff in order to solve challenges in the working area.

What is the most likely management skill applied in this case?

- A. Interpersonal skill
- B. Conceptual skill
- C. Operational skill
- D. Technical skill

ANSWER: A

159. A new graduate of medical radiology technologist is hired in a hospital and has started work in the department. The unit manager of the hospital informed the newly hired staff about expected activities from him, by giving guidance, ensuring adherence to rules and procedures, and regulating the work.

What is the most likely management function applied in this case?

- A. Directing
- B. Planning
- C. Organizing
- D. Controlling

ANSWER: A

160. A horrible car accident occurred, and multiple traumatic patients have been taken to an emergency department. The emergency unit manager made an immediate decision and gave directions to the health professionals to manage the mass casualty victims.

What is the most likely leadership style applied in this case?

- A. Participative
- B. Laissez faire
- C. Autocratic
- D. Motivative

ANSWER: C

161. A researcher planned to conduct a project in collaboration with concerned bodies to solve health-related problems in a selected community. The researcher has limited resources and selected only two problems based on the magnitude and severity of the problem, community concern, feasibility, and sustainability of the program.

What is the most likely planned step applied in this case?

- A. Situational analysis
- B. Identifying obstacles
- C. Designing the Strategies
- D. Priority setting

ANSWER: D

162. A primary hospital management team decided to develop a plan to improve the quality of health services. While planning, the team identified that a shortage of drug supply and inadequately trained human resources were the major obstacles to achieving the desired outcome.

What is the most likely strategic planning tool applied in this case?

- A. Threat
- B. Strength
- C. Weakness
- D. Opportunity

ANSWER: C

163. A radiology department head of a general hospital has developed a plan for unexpected problems and designed alternatives to the existing plan. The head is ready to use the plan in case the first plan proves inadequate due to changing circumstances.

What is the most likely type of plan developed by the head?

- A. Operational plan
- B. Contingency plan
- C. Tactical plan
- D. Strategic plan

ANSWER: B

164. A head of a woreda health office planned to form a health team that works on control and preventive measures of the outbreak of epidemic disease. In the course of team forming, conflicts among members were addressed and resolved, group unity formed, and the members developed an attitude of willingness to work and participate actively.

What is the most likely team development stage applied by the members?

- A. Forming
- B. Storming
- C. Norming
- D. Performing

ANSWER: C

165. A hospital manager decided to fill the vacant position of medical radiology department head by recruiting an experienced expert from another health organization. Through the acquisition process, the manager analyzed the knowledge and abilities needed for job success and measured the performance of the applicants, and chose a suitable applicant using the job qualification guide.

What is the most likely acquisition process applied by the manager in this case?

- A. Planning
- B. Selection
- C. Orientation
- D. Recruitment

ANSWER: B

166. A patient with a right leg fracture was presented to a radiology unit for leg x-ray imaging. The radiology technologist observed that the patient is in severe pain and told a nurse to apply pain management care before performing the X-ray imaging.

What is the most appropriate motivational theory applied in this case?

- A. Hierarchical Theory
- B. Two-factor theory
- C. Achievement theory
- D. Expectancy Theory

ANSWER: A

167. A referral hospital quality team has implemented an infection prevention and control program in the hospital in collaboration with multilevel health sector bodies. Throughout the project, the team continuously followed ongoing activities and discussed with health professionals and supervisors to check whether the work was being carried out as planned.

What is the most likely program controlling mechanism applied in this case?

- A. Evaluation
- B. Monitoring
- C. Organizing
- D. Audition

ANSWER: B

168. A hospital budget and project team implemented a cholera control and prevention program in a community. After completion of the project, evaluators wanted to assess the long-term effect of a program that meets the intended objective of the project.

What is the most effective type of evaluation to be applied in this case?

- A. Impact evaluation
- B. Process evaluation
- C. Formative evaluation
- D. Summative evaluation

ANSWER: A

169. A health extension worker who has been assigned to a village arranged a general discussion with the community to discuss personal and environmental hygiene. After discussing the issue, the extension worker told the members of the community to identify their problems, set priorities, make decisions, plan strategies, and implement them to achieve better health.

What is the most appropriate health promotion principle applied in this case?

- A. Health is not just an individual responsibility
- B. Health promotion works through community action
- C. Health promotion demands intersectoral collaboration
- D. Health promotion focuses on achieving equity in health

ANSWER: B

170. A senior radiologic technologist working in a hospital called ordinary meeting of all staff members of the department. After making an opening speech, the technologist requested everyone to identify their own needs for change and then select the best methods that will enable the department to improve its service delivery.

What is the most appropriate principle used in this case?

- A. Principle of segmentation
- B. Principle of participation
- C. Principle of need-based
- D. Principle of culture

ANSWER: B

171. A health education instructor planned to teach students about the importance and levels of health education in disease prevention. During the lecture, the instructor emphasized educating sick people to seek treatment, breast-cancer screening, blood pressure examination, cholesterol level examination, and treating malaria patients.

What is the most appropriate level of disease prevention method emphasized in this case?

- A. Primary
- B. Tertiary
- C. Secondary
- D. Primordial

ANSWER: C

172. A new radiology technology graduate entered an X-ray machine room and disliked it because it was small. Likewise, another new graduate found the same room to be attractive.

What is the most likely factor affecting the behavior of the new graduates in this case?

- A. Value
- B. Belief
- C. Attitude
- D. Knowledge

ANSWER: C

173. Breast self-examination and mammography campaigns often identify people who are at high risk for breast cancer, but who say they have not experienced any symptoms. As to the health belief model, print materials, reminder letters, or pill calendars might encourage people to consistently follow their doctors' recommendations.

What is the most appropriate construct of the health belief model recommended?

- A. Self-efficacy
- B. Cues to action
- C. Perceived barriers
- D. Perceived benefits

ANSWER: B

174. A senior radiologic technologist wanted to assess the practice of using lead aprons among radiation workers during pelvic x-ray imaging. From the assessment, the technologist found that most of the radiation workers were not using the radiation protection device. From this result, the technologist designed potential change strategies to promote behavior change, such as making adjustments to the environment or influencing personal traits.

What is the most likely theory addressed in this case?

- A. Health belief model
- B. Transtheoretical model
- C. Social Cognitive Theory
- D. Diffusion of Innovations

ANSWER: C

175. A radiology technology instructor planned to teach year-four students about anatomical parts seen on chest x-ray. The radiologic technology instructor brought radiographic images and distributed them to each student so that they could see the anatomical parts that are seen on the image.

What is the most appropriate educational aid used in this case?

- A. Audio
- B. Visual
- C. Health talk
- D. Audio-visual

ANSWER: B

176. An instructor appointed a group of radiologic technology students who will start ultrasound hospital attachment by the coming week. Then, the instructor showed the students step-by-step techniques of ultrasound scanning of the liver. Then, the students practiced the scanning techniques turn by turn.

What is the most likely teaching method used by the instructor?

- A. Drama
- B. Lecture
- C. Counselling
- D. Demonstration

ANSWER: D

177. A 21-year-old female patient was brought to a radiology unit for chest x-ray examination. The radiology technologist informed the patient to remove her clothes above the waist as part of preparation for the examination. However, the patient refused because she claimed that she did not expose her body to a male person. As a result, the radiology technology accepted the patients' decision and planned another alternative.

What is the most likely ethical principle revealed in this case?

- A. Justice
- B. Autonomy
- C. Beneficence
- D. Non-maleficence

ANSWER: B

178. While performing an ultrasound examination, a patient with emergency case of ectopic pregnancy presented to a radiology unit. Consequently, the radiology technologist instructed the patient to expose her body and prepare for the examination while another patient was in the same room.

What is the most likely ethical principle violated in this case?

- A. Consent
- B. Privacy
- C. Curiosity
- D. Communication

ANSWER: B

179. A radiology unit at a rural hospital ran out of a radiographic film processing chemical. As a result, a radiology unit head decided to withhold rendering radiology services for routine radiographic examination and utilized the few remaining inputs only for emergency services. However, the large population waiting for radiology services has complained about the delay.

What is the most likely health care dilemma the radiology unit encountered in this case?

- A. End-of-life management
- B. Disclosure of supply
- C. Resource scarcity
- D. Refusal to treatment

ANSWER: C

180. A radiology technologist at a private clinic allowed a senior radiographer to cover his duty since he was on a week off for a holiday. While performing an ultrasound examination, the radiographer misdiagnosed a twin pregnancy. As a result, the patient came to the clinic to address the matter and showed her disappointment.

What is the most likely ethical liability revealed in this case?

- A. Duty to fully record personal health information.
- B. Duty to maintain personal conduct and integrity.
- C. Duty to practice within the scope of professional practice.
- D. Duty to report the existence of professional malpractice.

ANSWER: C

181. An anatomy lecturer brought radiology technology students to a cadaver room for a skill laboratory demonstration. It was a first experience for the students to see a dead human body. They were shocked to watch the senior students practicing on the dead body. However, the students later realized that it was part of medical students' life and came to terms with the issue and continued the demonstration.

What is the most likely principle of compassionate care demonstrated in this case?

- A. Affection
- B. Attention
- C. Acceptance
- D. Acknowledgement

ANSWER: C

182. While positioning a 70-year-old female patient to perform ultrasound examination, a radiology technologist shouted at the patient and instructed her to remove her clothes quickly. In addition, the technologist informed the patient that he will let another patient in instead if she does not get prepared soon.

What is the most likely threat to compassionate care demonstrated in this case?

- A. Non-dignified care
- B. Non-consented care
- C. Non-confidential care
- D. Non-condemning care

ANSWER: A

183. While a junior technologist was performing an X-ray examination of a 40-year-old patient, the radiology unit head suddenly entered the examination room. The unit head angrily shouted at the technologist and informed him that he was using inappropriate radiographic projection.

What is the most likely aspect of compassionate and motivated care breached in this case?

- A. Non-verbal communication
- B. Co-worker communication
- C. Professional-client communication
- D. Professional-family communication

ANSWER: B

184. A six-year-old schoolboy was brought to a radiology unit for x-ray examination. The patient was frightened and anxious.

What is the most effective communication strategy to be applied in this case?

- A. Intimidating the patient to follow instructions.
- B. Providing the patient with simple written instructions.
- C. Involving the patient in decision about the procedure.
- D. Using a model to show the patient the procedure.

ANSWER: D

185. While a radiology technologist was performing a routine x-ray examination, a bedridden patient with a recent car accident was brought to a radiology unit for X-ray examination.

What is the most appropriate initial step the technologist should follow?

- A. Securing informed consent
- B. Ensuring radiation protection
- C. Reviewing the request form
- D. Performing radiographic projection

ANSWER: C

186. A physician wants to compare the progress of the patient who has been on anti-TB treatment for about six months. Accordingly, the physician requests the radiology department a radiographic image of the patient. However, the radiographic images are not retrieved since the images saved on the computer lack the appropriate patient identifier information.

What is the most likely malpractice described in this case?

- A. Courtesy
- B. Charting
- C. Compliance
- D. Communication

ANSWER: B

187. A community health worker wants to conduct a Community Health Assessment to identify the major problems of the community in a comprehensive and stepwise approach.

What is the first procedure that should be taken in this case?

- A. Engage the community and understand their health priorities
- B. Identify key partners and stakeholders
- C. Develop a community health improvement plan
- D. Describe the community and the population

ANSWER: D

188. A woreda health department organized a campaign for cervical cancer screening for two consecutive fiscal years (2023 and 2024). A total of 20,000 reproductive age women took part in the campaign, 10,000 in each year. It was found that 380 women in 2023 and 220 women in 2024 were positive for cervical cancer.

What is the period prevalence in the above scenario?

- A. 30 per 1000 women
- B. 19 per 1000 women
- C. 11 per 1000 women
- D. 60 per 1000 women

ANSWER: A

189. A specific woreda in Ethiopia has a total of 100,000 reproductive aged and 12,433 menopause women. In the same fiscal year, the district health office reported 50,000 live births, 1000 neonatal deaths, and 264 maternal deaths.

What is the maternal mortality rate of the woreda in this case?

- A. 528 per 100,000
- B. 264 per 100,000
- C. 176 per 100,000
- D. 517.6 per 100,000

ANSWER: A

190. A radiologist wants to measure the level of satisfaction among patients who received radiation therapy. The radiologist intends to report on the satisfaction level disaggregated by different characteristics of the study participants, such as age, residence status, and marital status.

What is the most likely type of research applied in this case?

- A. Exploratory
- B. Analytic
- C. Descriptive
- D. Correlational study

ANSWER: C

191. A team of radiologists conducted a study aimed at assessing the burden of femoral fractures among construction workers. In addition, they also want to identify the potential influences related to femoral fractures.

What is the most appropriate general objective for this case?

- A. Determine the magnitude of femoral fracture and its associated factors
- B. Study the prevalence of femoral fractures and their associated factors
- C. Find out factors associated with femoral fracture
- D. Appreciate the magnitude of femoral fracture and its determinants

ANSWER: A

192. A national AIDS Control Program wants to establish counselling services for HIV/AIDS patients. However, to provide such support, the office did not have adequate information on the specific needs of the patients. To assess these needs, some in-depth interviews are held by taking various categories of patients (males and females) and counsellors who are working on the program.

What is the most appropriate study design for this case?

- A. Explanatory
- B. Comparative
- C. Exploratory
- D. Longitudinal

ANSWER: C

193. A researcher wants to identify the potential risk factors of bed ulcers among patients admitted to the Intensive Care Unit of a hospital. In the process, the researcher selects patients with and without bed ulcers. The exposure to possible risk factors is then determined and compared in both groups independently.

What is the most appropriate study design to be employed in this case?

- A. A comparative cross-sectional study
- B. Case-control study
- C. Hospital-based survey
- D. Retrospective cohort study

ANSWER: B

194. A researcher wants to conduct a study to assess the magnitude of anemia among pregnant women who visit antenatal care clinic. There was a total of 3000 pregnant women who visited the clinic at the time of the study. The researcher decided to give an equal chance of selection for all pregnant women to be included in the study. Finally, the researcher took a sample of 422 pregnant women from the existing lists of records of pregnant women in the ANC clinic.

What is the most likely sampling technique applied for this case?

- A. Stratified sampling
- B. Cluster sampling
- C. Convenience sampling
- D. Simple random sampling

ANSWER: D

195. A professor wants to explore the potential barriers to a conducive teaching-learning environment at higher education institutions. Due to limited time and resources, the professor decides to interview all his colleagues who are invited to attend the research conference at one of the universities in the country.

What is the most appropriate sampling technique applied in this case?

- A. Non-random sampling
- B. Haphazard sampling
- C. Convenience sampling
- D. Quota sampling

ANSWER: B

196. A researcher wants to conduct a study to determine the magnitude of femoral fracture among patients who visited the orthopedic department of a given referral hospital. There was no prior study conducted in the area of femoral fracture. The researcher assumes a degree of accuracy of four percent, a confidence level of 95 percent, and a non-response rate of 10 percent.

What is the maximum sample size required for this specific study?

- A. 384
- B. 423
- C. 601
- D. 661

ANSWER: D

197. A Midwife wants to measure the prevalence of long-term contraceptive service uptake among women of reproductive age groups who visited a family planning clinic. One of the questions embedded in the questionnaire deals with the marital status of the women.

What is the most appropriate questioning type fit for assessing marital status in this case?

- A. Close-ended
- B. Open-ended
- C. Self-administered
- D. Mixed

ANSWER: A

198. A researcher wants to conduct a study on the magnitude of sexual violence and its associated factors among 150 female university students. The researcher, who is the staff of the same university, planned to gather all the participants in one place at one time, giving oral instructions and letting them fill out the questionnaires.

What is the most appropriate data collection method to be applied in this case?

- A. Interviewer-administered interview
- B. Mailed interview
- C. Self-administered interview
- D. Observation

ANSWER: C

199. A radiologist wants to categorize the stages of breast cancer following a mammography examination for a total of 135 women. Accordingly, 60, 20, 20, and 35 of the women were categorized as stage I, II, III, and IV, respectively.

What is the most likely category of variable reported in this case?

- A. Quantitative
- B. Qualitative
- C. Numerical discrete
- D. Interval

ANSWER: B

200. A hospital manager wants to present the trends of cases diagnosed with malaria for the past ten years in a community. The manager obtained the malaria cases reported weekly from a laboratory registration book of a nearby health post.

What is the most appropriate graphical presentation fit for this case?

- A. Line diagram
- B. Frequency polygon
- C. Histograms
- D. Bar graph

ANSWER: A